

QUESTION 21

Build an End-to-End RAG Pipeline

Document Loading • Text Chunking • Embeddings • Retrieval • Answer Generation

FAISS + all-MiniLM-L6-v2 + google/flan-t5-small

1. Aim

To build an end-to-end Retrieval-Augmented Generation pipeline that loads a document, divides it into text chunks, generates embeddings, stores the embeddings in a vector store, retrieves relevant chunks for a question, forms context, and generates a final answer.

2. Problem Statement

A complete RAG application requires more than a retrieval index. Raw documents must first be loaded and divided into manageable chunks. Those chunks must be converted into vectors and indexed. At query time, relevant chunks must be retrieved and supplied to a generation model as context. This practical implements the complete workflow in one pipeline.

3. Objectives

- Load information from a source document.
- Split the document into meaningful text chunks.
- Generate embeddings for each chunk.
- Store chunk embeddings in a FAISS vector store.
- Convert the user question into an embedding.
- Retrieve the Top-K relevant chunks.
- Combine retrieved chunks into a context.
- Generate an answer using FLAN-T5.

4. End-to-End Architecture

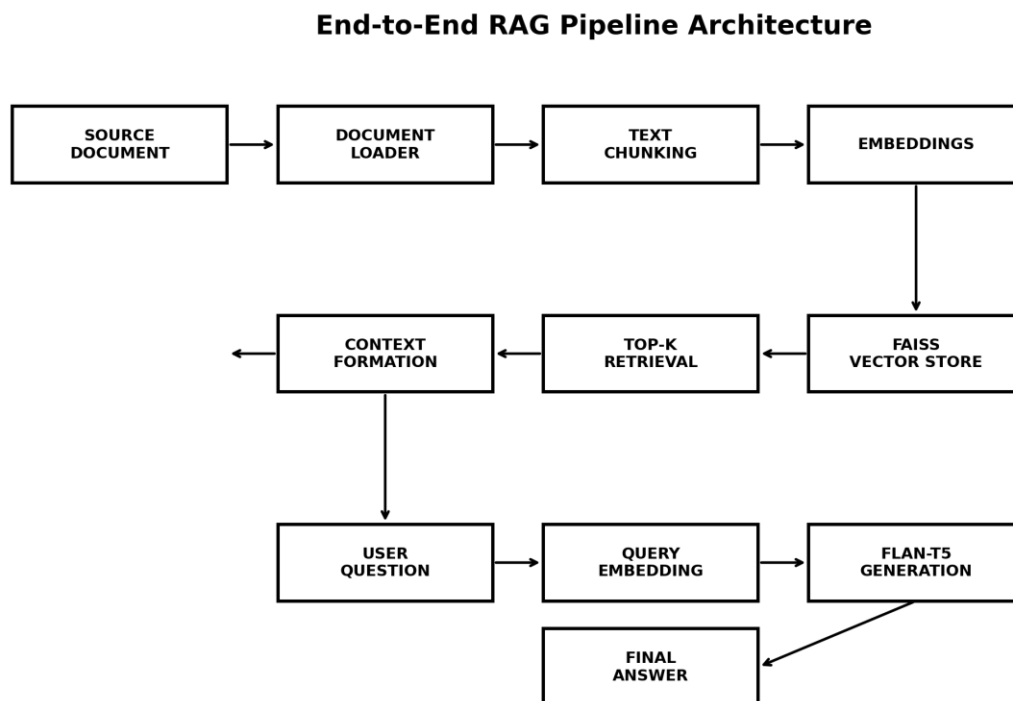


Figure 1: Complete end-to-end RAG pipeline implemented in Question 6.

5. Pipeline Stages

Stage	Function
Document Loading	The source file knowledge.txt is read into the application.
Text Chunking	The loaded text is divided into smaller chunks for efficient retrieval.
Embedding Generation	Each chunk is converted into a 384-dimensional vector using all-MiniLM-L6-v2.
Vector Storage	Chunk vectors are normalized and stored in a FAISS IndexFlatIP index.
Query Processing	The user's question is converted into a vector using the same embedding model.
Retrieval	FAISS searches the vector store and returns the three most relevant chunks.
Context Formation	The retrieved chunks are combined into a single context.
Answer Generation	google/flan-t5-small generates an answer from the question and retrieved context.

6. Implementation Configuration

Parameter	Actual Value
Source Document	knowledge.txt
Text Chunks	5
Embedding Model	all-MiniLM-L6-v2
Embedding Dimension	384
Vector Store	FAISS
Index Type	IndexFlatIP
Retrieved Chunks	3
Generation Model	google/flan-t5-small

7. Python Implementation

```
# Load source document
with open("knowledge.txt", "r", encoding="utf-8") as file:
    text = file.read()

# Split document into chunks
paragraphs = [
    p.strip() for p in text.split("\n\n") if p.strip()
]

chunks = []
for paragraph in paragraphs:
    words = paragraph.split()
    for i in range(0, len(words), 45):
        chunk = " ".join(words[i:i + 45])
        if chunk:
            chunks.append(chunk)
```

```

# Generate chunk embeddings
chunk_embeddings = embedding_model.encode(
    chunks, convert_to_numpy=True
).astype("float32")

faiss.normalize_L2(chunk_embeddings)

# Create FAISS vector store
vector_store = faiss.IndexFlatIP(
    chunk_embeddings.shape[1]
)
vector_store.add(chunk_embeddings)

# Retrieve Top-K chunks
query_embedding = embedding_model.encode(
    [question], convert_to_numpy=True
).astype("float32")
faiss.normalize_L2(query_embedding)

scores, indices = vector_store.search(
    query_embedding, 3
)

# Build context
context = "\n\n".join(
    chunks[int(i)] for i in indices[0]
)

# Generate answer
prompt = f"""Answer the question using only the information
provided in the context.

Context:
{context}

Question:
{question}

Give a short and clear answer.

Answer:"""

inputs = tokenizer(
    prompt, return_tensors="pt",
    truncation=True, max_length=512
)

outputs = generator.generate(
    **inputs, max_new_tokens=100,
    do_sample=False
)

```

```
answer = tokenizer.decode(  
    outputs[0], skip_special_tokens=True  
)
```

8. Test Question

What is machine learning?

9. Actual Retrieval Output

```
END-TO-END RAG PIPELINE  
  
Question: What is machine learning?  
  
RETRIEVED TEXT CHUNKS  
  
Rank 1 | Chunk ID: 0 | Score: 0.8465  
Machine learning is a branch of artificial intelligence.  
It enables computers to learn patterns from data without  
being explicitly programmed for every task.  
  
Rank 2 | Chunk ID: 1 | Score: 0.6844  
Machine learning can be divided into supervised learning,  
unsupervised learning, and reinforcement learning.  
  
Rank 3 | Chunk ID: 4 | Score: 0.5331  
Artificial intelligence combines different computational  
methods to create systems capable of learning, reasoning,  
problem solving, and decision making.  
  
CONTEXT FORMATION  
Context created successfully.  
Context length: 531  
  
FINAL GENERATED ANSWER  
a branch of artificial intelligence  
  
PIPELINE INFORMATION  
Source Document      : knowledge.txt  
Text Chunks          : 5  
Embedding Model       : all-MiniLM-L6-v2  
Embedding Dimension   : 384  
Vector Store          : FAISS  
Index Type            : IndexFlatIP  
Retrieved Chunks      : 3  
Generation Model       : google/flan-t5-small  
  
END-TO-END RAG PIPELINE COMPLETED SUCCESSFULLY
```

Figure 2: Actual end-to-end RAG output obtained from the implementation.

10. Retrieved Chunk Results

Rank	Chunk ID	Similarity Score	Retrieved Chunk
1	0	0.8465	Machine learning is a branch of artificial intelligence. It enables computers to learn patterns from data without being explicitly programmed for every task.
2	1	0.6844	Machine learning can be divided into supervised learning, unsupervised learning, and reinforcement learning. Supervised

			learning uses labelled data for training. Unsupervised learning discovers hidden patterns in data.
3	4	0.5331	Artificial intelligence combines different computational methods to create systems capable of learning, reasoning, problem solving, and decision making.

11. Retrieval Score Visualization

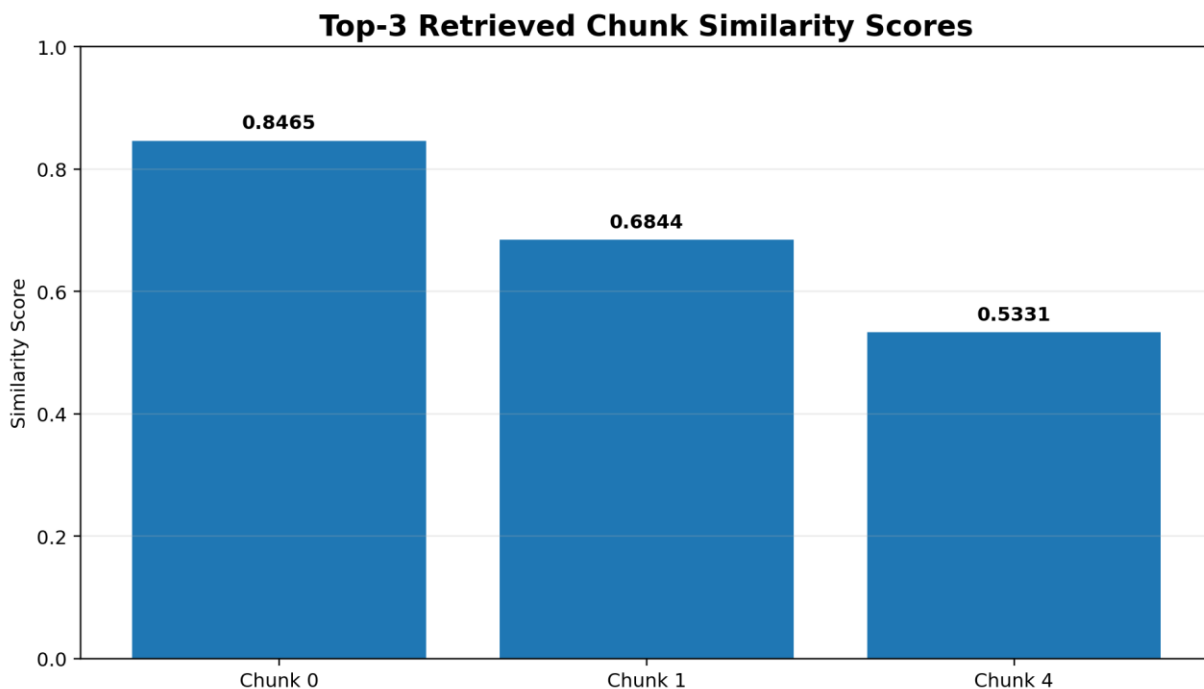


Figure 3: Similarity scores of the three chunks retrieved from the vector store.

12. Context Formation

After retrieval, the three selected chunks were combined into a single context. The actual context length reported by the implementation was 531 characters. This context was passed to the generation model together with the question.

13. Generated Answer

Actual generated answer: “a branch of artificial intelligence”

The generated answer is concise and is directly supported by the highest-ranked chunk, which states that machine learning is a branch of artificial intelligence.

14. Result Analysis

The end-to-end pipeline successfully loaded knowledge.txt, created 5 text chunks, generated 384-dimensional embeddings, stored them in FAISS, retrieved the top 3 chunks, formed a context of 531 characters, and generated an answer using FLAN-T5. The highest similarity score was 0.8465 for Chunk 0, which contained the most directly relevant definition.

15. Advantages

- Demonstrates the complete RAG workflow in one application.
- Supports document-based retrieval rather than fixed question-answer pairs.
- Chunking makes large documents easier to search.
- Vector retrieval selects relevant context before generation.
- The source document can be replaced without changing the overall pipeline.

16. Applications

- Document question answering
- Knowledge-base assistants
- Research document search
- Educational assistants
- Technical documentation assistants
- Enterprise RAG systems

17. Conclusion

An end-to-end Retrieval-Augmented Generation pipeline was successfully implemented. The system loaded knowledge.txt, divided the content into 5 chunks, generated 384-dimensional embeddings, indexed the chunks using FAISS, retrieved 3 relevant chunks for the question, formed a 531-character context, and generated the final answer using google/flan-t5-small. The highest retrieval score was 0.8465, confirming successful semantic retrieval and context selection.